Paper 4.75 - Boundary Repair as Next-State Precondition

CQE-paper-04.75

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-04.75.md

Paper 4.75 - Boundary Repair as Next-State Precondition

Purpose

Paper 4.75 explains how boundary repair becomes the precondition for Paper 5's path carrier and for later causal-code review.

Exported State

Paper 4 exports:

```
repaired constraint rows
original LCR states
Paper 3 axis/sheet coordinates
reason: Paper 2 correction fired
status: constraint
next legal routes
source and target paper labels
```

Use in Paper 5

Paper 5 may attach a Paper 4 repair row as a payload only if the payload does not alter the path rule.

The allowed transition is:

```
repair_boundary(s) = constraint row r
attach_payload(q_t, r) = carried receipt at path state q_t
payload_alters_path_rule = false
```

Paper 5 proves the path-carrier rule. Paper 4 supplies the payload object.

Use in Paper 6

Paper 6 may register Paper 4 repair rows as typed causal edges. The edge must name its source, target, edge type, receipt, and status. Paper 4 proves the row; Paper 6 proves the dependency graph discipline.

Use in Later Papers

Later papers may lift boundary repair into:

- rolling path carriers,
- causal proof graphs,
- curvature and geometry repair language,
- mass-residue or Hamiltonian accumulated-center claims,
- product and adapter workflows.

Every lift must state which later theorem closes the added structure.

Precondition Rule

Before a later paper uses Paper 4, it should be able to answer:

```
What failed?
What state and coordinate were preserved?
```

```
Why is the row a constraint rather than proof?
Which next route consumes it?
Which receipt proves the repair row?
```

Conclusion

Paper 4.75 turns repair into portable state. It exports failure as structured constraint so Paper 5 can carry it and Paper 6 can audit it.